

Isolation and Structures of Two New Organozinc Anions from Solutions Rich in Halide Ions

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The organometallic dianions diethyltetrachlorodizincate, $[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$, and ethyltribromozincate, $[\text{EtZnBr}_3]$, have been isolated from reaction mixtures containing diethylzinc and a tetraphenylphosphonium halide ($\text{X} = \text{Cl}$ or Br). Evidence for these anions had previously been reported using mass spectrometry. The ions may have importance in the solution structure of organozinc reagents used extensively in organic synthesis.

Introduction

Organometallic reagents are essential for modern organic synthesis, but are often poorly understood in terms of both mechanism and solution speciation. Subtle effects of the solvent and dissolved ions can play an important role in determining which products will be formed in a given reaction.¹ This can lead to varying yields, unexpected products, and oftentimes unpredictable results of reactions.² Organozinc reagents have recently seen a resurgence in popularity due to their potential as sources of functionalized carbanions.³ They are structurally diverse, and their reactivity is often controlled by many factors including the substituents, ligands, and, of course, the solvent.

The reactivity of organozinc reagents is broadly governed by charge, with diorganozincs being the least reactive, organozinc halides being more so, and triorganozincate anions being the most reactive.³ The isolation of doubly charged anionic organozinc compounds described in the current paper suggests a further class of potentially highly reactive organozinc reagents. To support this, earlier reports have indicated that organozinc complexes with a formal 2-charge can exist in certain situations.⁴ In a two-part study by Koszinowski and Böhrer,^{5,6} anion-mode electrospray ionization mass spectrometry was used to identify a variety of ZnR_n^- and LiZnR_n^- species ($\text{R} = \text{X}$, alkyl). In this study, a 10 mM solution of $\text{LiBu}/\text{ZnCl}_2$ displayed high concentrations of $\text{LiZn}_2\text{Bu}_2\text{Cl}_4^-$ and $\text{Zn}_2\text{BuCl}_4^-$ ions, and further it was found that increasing the concentration of the solution

rendered these the dominant species. We report herein the structural characterization of similar highly charged anionic organozinc species.

Our group has been interested in the use of basic organometallic reagents in ionic liquids (ILs) for some time, and we have identified interesting reactivity for such reagents.⁷ Specifically, we have reported results in which a phosphonium ionic liquid was used to shift the Schlenk equilibrium of a Grignard reagent, and we characterized different modes of reactivity in ionic and molecular solvents. For example, it was observed that a solution of RMgBr ($\text{R} = \text{Et}$, Ph , Mes) and *p*-benzoquinone in a phosphonium-based ionic liquid reacted via an electron transfer process, giving *p*-hydroxyphenol as the major product, rather than 4-hydroxy-4-alkylcyclohexa-2,5-dienone, as was observed in THF. In the phosphonium ionic liquids utilized, we were unable to prepare the Grignard reagent directly from a combination of magnesium metal and an alkyl or aryl halide, but we note that they have been prepared in ILs by other groups.⁸

Charged reagents have a significant advantage in reactivity over their noncharged equivalents. Opposing charges, even partial or temporary charges, draw species together and can promote reaction. Various groups, particularly that of Knochel, have shown that the addition of coordinating ions to reactions of organometallic compounds, and alkali metal halides ions (most notably LiCl) can result in enhanced reactivity of the organometallic reagent.^{9,10} Ionic liquids, being solutions composed of *discrete ions*, have the ability to alter the reactivity of many species through the stabilization of reactive

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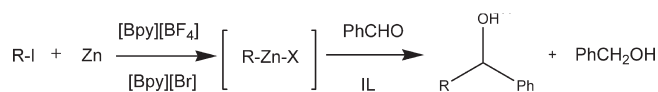
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Scheme 1. Preparation and Reactivity of Organozinc Reagents in an *N*-Butylpyridinium Ionic Liquid with an External Halide Source



ions and intermediates formed during the reaction. As seen in the synthetic report by Law, Wong, and Chan, unexpected reactivity was observed in a solution of an organozinc reagent in ionic liquid media when bromide ions were added to the solution.⁸ In this report Chan⁸ noted a dramatic shift from the typical β -hydride reduction of aldehydes by organozinc reagents toward a pathway yielding primarily the alkylated product (Scheme 1), and the authors speculated on the presence of a novel organometallic species in this highly ionic media. Although organometallic reagents typically coordinate with anionic ligands (ions) in ionic liquids, it is important to note that ILs are two-component mixtures (cation–anion pairs). We thus cannot discount the possibility of a reducing species (a hydride source) being formed through some reaction involving the cation and the organometallic species.

Of particular importance was their observation that the yield of this reaction was heavily dependent upon the bromide concentration of the solution. Thus we may infer that halide ions are particularly capable of influencing the speciation and reactivity of organozinc reagents, as they have been shown to be in the chemistry of other organometallics.^{11,12} Finally, it is important to note here the careful NMR study by Richey that examined the activation of dialkylzinc compounds with a variety of ammonium and alkali metal salts. These studies clearly showed formation of active species, but they were unable to definitively assign structures for the complexes being formed.¹³

Experimental Section

An argon atmosphere double-manifold vacuum line and argon atmosphere drybox (mBraun Unilab) were used for the manipulation of air- and moisture-sensitive compounds. Ultra-high-purity argon gas was used in all cases. Anhydrous solvents were available directly using an mBraun MB-SPS solvent purification system. All melting points were determined using a Barnstead Mel-Temp apparatus. NMR spectra were obtained using a Bruker Avance 500 MHz spectrophotometer. IR spectra were obtained using a Bruker Vertex 70 infrared spectrometer as either potassium bromide pellets (for moisture-stable compounds) or dichloromethane solutions between potassium bromide plates.

The NMR spectra were obtained at the Nuclear Magnetic Resonance Research Resource (NMR-3) at Dalhousie University, Halifax, Canada. ¹H and ¹³C spectral shifts were reported in relation to either known residue solvent peaks or TMS, when available. ³¹P spectral shifts were taken without internal calibration. NMR spectra were referenced to the solvent, and chemical shifts are reported in ppm relative to an external standard (TMS for ¹H and ¹³C, 85% H₃PO₄ for ³¹P). Elemental analyses were performed by the Canadian Microanalytical Service, Delta, British Columbia. Data for [PPh₄]₂[EtZnBr₃] were collected by Dr. Gabrielle Schatte at The Saskatchewan Structural Sciences Centre (SSSC).

Preparation of [PPh₄]₂[Et₂Cl₄Zn₂]. Tetraphenylphosphonium chloride (375 mg) was weighed and added into a predried 50 mL Schlenk flask. The flask was sealed and evacuated for ~1 h. The flask was brought into the drybox and opened, and a stir bar was added. The salt was dissolved in ca. 10 mL of CH₂Cl₂, forming a clear, colorless solution. Cold ZnEt₂ solution (1 M in hexanes, 1.2 mL) was then added dropwise. After addition of the ZnEt₂ solution, the reaction took on a faint yellow tinge, which darkened slightly over the next 10 min. The solution was set aside to evaporate to dryness in the reaction flask. Overnight the solution evaporated completely, producing small clear-yellow crystals. The dry crystals were washed with two washings each of hexanes, toluene, and THF. These crystals were then characterized as [PPh₄]₂[Et₂Cl₄Zn₂]. Yield = 0.11 g, 22%, mp = 227–230 °C. ¹H NMR (CD₂Cl₂): δ 7.92 (m, 8H), 7.76 (m, 16H), 7.72 (m, 16H), 1.05 (t, 6H), –0.11 (q, 4H). Anal. Calcd for C₅₂H₅₀Cl₄P₂Zn₂: C, 61.87; H, 4.99; N, 0; Cl, 14.05. Found: C, 59.26; H, 4.44; N, <0.3; Cl, 20.90 (note: this sample appears to be solvent sensitive and absorbs 1 equiv of dichloromethane from the drybox atmosphere; the EA data are high for chloride ion, but the NMR spectrum obtained for the sample used for the EA experiments contained 1 equiv of CH₂Cl₂, as determined by integration; thus, C₅₂H₅₀Cl₄P₂Zn₂·CH₂Cl₂ has a calculated EA of C, 58.16; H, 4.79; Cl, 19.44).

Preparation of [PPh₄]₂[EtZnBr₃]. Tetraphenylphosphonium bromide (420 mg) was weighed and added into a predried 50 mL Schlenk flask. The flask was sealed and evacuated for ~1 h. The flask was brought into the drybox and opened, and a stir bar was added. The salt was dissolved in ~10 mL of CH₂Cl₂, forming a clear, colorless solution. Cold ZnEt₂ (–34 °C) solution (1 M in hexanes), totaling 1.2 mL, was added dropwise to the reaction vessel over a period of 1 min. Immediately after addition of the metal solution, the reaction took on a slight yellow tinge, which deepened in color over the next 10 min. After 1 h of stirring, the solution was transferred by pipet into a 20 mL screw cap vial. Overnight the solution evaporated to ~2 mL in volume, producing clear-yellow crystals in a now deep red-brown solution. Some of these crystals were removed and put into a separate vial, while others were left in the mother solution. In both cases, the crystal-containing vials were sealed to prevent further evaporation. The dry crystals were washed with two portions each of hexanes, toluene, and THF. These crystals were then characterized as [PPh₄]₂[EtZnBr₃] using melting point, infrared, NMR, and elemental analysis data. Yield = 0.21 g, 42%, based upon [Ph₄P][Br] as the limiting reagent; mp = 242–245 °C. ¹H NMR (DMSO-*d*₆): δ 7.97 (m, 8H), 7.81 (m, 16H), 7.74 (m, 16H), 1.03 (t, 2H), –0.20 (q, 3H). Anal. Calcd for C₅₀H₄₅Br₃P₂Zn: C, 59.29; H, 4.48; N, 0; Cl, 0. Found: C, 58.54; H, 4.44; N, >0.3; Cl, >0.3.

Preparation of [PPh₄]₂[ZnCl₄]. Tetraphenylphosphonium chloride (1.00 g) was weighed out in air and added to a 250 mL round-bottom flask. Then 370 mg of zinc(II) chloride was added, and the mixture was dissolved in THF. The solution was heated to ~66 °C, with attached reflux condenser, for 10 h. On completion, the solution was allowed to evaporate over several days, finally producing yellowish crystals. Yield = 0.57 g (42%), mp = 179–182 °C.

Preparation of [PPh₄]₂[ZnBr₄]. Tetraphenylphosphonium bromide (1.00 g) was weighed out in air and added to a 250 mL round-bottom flask. Then 540 mg of zinc(II) bromide was added, and the mixture was dissolved in THF. The solution was heated to ~66 °C, with attached reflux condenser, for 10 h. On completion, the solution was allowed to evaporate over several days, finally producing colorless crystals. Yield = 1.09 g (71%), decomposition range 184–195 °C.

Crystallography. Data for [PPh₄]₂[EtZnBr₃] were collected on a Nonius Kappa CCD 4-Circle Kappa FR540C diffractometer using monochromated Mo K α radiation (λ = 0.71073 Å) at the

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Table 1. Summary of Crystal Data and Structure Refinement^a

identification code	[Ph ₄ P] ₂ [Et(Cl)Zn-(μ -Cl) ₂ Zn(Cl)Et]	[Ph ₄ P] ₂ [EtZnBr ₃]	[Ph ₄ P] ₂ [ZnBr ₄]	[Ph ₄ P] ₂ [ZnCl ₄]	[Ph ₄ P] ₂ [Cl ₂ Zn-(μ -Cl) ₂ ZnCl ₂]
empirical formula	C _{105.40} H _{103.44} Cl _{17.32} P ₄ Zn ₄	C _{49.77} H _{44.42} Br _{3.12} P ₂ Zn	C ₄₈ H ₄₀ Br ₄ P ₂ Zn	C ₄₈ H ₄₀ Cl ₄ P ₂ Zn	C ₄₈ H ₄₀ Cl ₁₆ P ₂ Zn ₂
fw	2014.4	1018.79	1063.75	885.91	
temperature	296(2) K	173(2) K	296(2) K	296(2) K	
wavelength	0.71073 Å	0.71073 Å	0.71073 Å	0.71073 Å	
cryst syst	triclinic	orthorhombic	monoclinic	monoclinic	triclinic
space group	<i>P</i> $\bar{1}$	<i>Pbca</i>	<i>C2/c</i>	<i>C2/c</i>	<i>P</i> $\bar{1}$
unit cell dimensions					
<i>a</i>	11.4075(11) Å	17.3180(4) Å	11.182(3) Å	10.9888(11) Å	9.736(1) Å
<i>b</i>	13.3666(13) Å	20.9220(3) Å	19.633(6) Å	19.491(3) Å	9.826(2) Å
<i>c</i>	17.1784(16) Å	24.8930(5) Å	20.531(5) Å	20.269(2) Å	12.967(3) Å
α	86.4880(10)°	90°	90°	90°	107.93(2)°
β	71.1050(10)°	90°	91.802(3)°	91.6220(10)°	94.50(2)°
γ	87.1710(10)°	90°	90°	90°	100.91(2)°
volume	2472.3(4) Å ³	9019.4(3) Å ³	4505(2) Å ³	4339.5(8) Å ³	1146.443 Å ³
<i>Z</i>	1	8	4	4	1
density (calcd)	1.353 Mg/m ³	1.501 Mg/m ³	1.568 Mg/m ³	1.356 Mg/m ³	
abs coeff	1.267 mm ⁻¹	3.413 mm ⁻¹	4.196 mm ⁻¹	0.919 mm ⁻¹	
<i>F</i> (000)	1040	4097	2112	1824	
cryst size	0.460 × 0.421 × 0.240 mm ³	0.30 × 0.25 × 0.25 mm ³	0.42 × 0.35 × 0.31 mm ³	0.25 × 0.21 × 0.16 mm ³	
θ range for data collection	2.02 to 28.57°	2.89 to 27.48°	2.07 to 25.00°	2.09 to 25.00°	
index ranges	-15 ≤ <i>h</i> ≤ 15 -17 ≤ <i>k</i> ≤ 16 -23 ≤ <i>l</i> ≤ 22	-22 ≤ <i>h</i> ≤ 22 -26 ≤ <i>k</i> ≤ 27 -32 ≤ <i>l</i> ≤ 32	-13 ≤ <i>h</i> ≤ 13 -23 ≤ <i>k</i> ≤ 23 -24 ≤ <i>l</i> ≤ 23	-13 ≤ <i>h</i> ≤ 13 -23 ≤ <i>k</i> ≤ 23 -22 ≤ <i>l</i> ≤ 24	
reflns collected	20 288	18 862	14 926	14 306	
indep reflns	11 332 [<i>R</i> (int) = 0.0186]	10 305 [<i>R</i> (int) = 0.0548]	3938 [<i>R</i> (int) = 0.0184]	3818 [<i>R</i> (int) = 0.0249]	
completeness to $\theta = 25.00^\circ$	97.7%	99.5%	98.9%	99.8%	
absorp corr		semiempirical from equivalents			
max. and min. transmn	0.7457 and 0.5573	0.4824 and 0.4275	0.7458 and 0.6067	0.7458 and 0.6557	
refinement method		full-matrix least-squares on <i>F</i> ²			
data/restraints/params	11 332/18/589	10 305/6/524	3938/0/249	3818/0/249	
goodness-of-fit on <i>F</i> ²	1.015	1.048	1.266	1.034	
final <i>R</i> indices	<i>R</i> 1 = 0.0432, w <i>R</i> 2 = 0.1034	<i>R</i> 1 = 0.0618, w <i>R</i> 2 = 0.1275	<i>R</i> 1 = 0.0600, w <i>R</i> 2 = 0.2112	<i>R</i> 1 = 0.0367, w <i>R</i> 2 = 0.0959	
<i>R</i> indices (all data)	<i>R</i> 1 = 0.0806, w <i>R</i> 2 = 0.1198	<i>R</i> 1 = 0.1140, w <i>R</i> 2 = 0.1478	<i>R</i> 1 = 0.0706, w <i>R</i> 2 = 0.2235	<i>R</i> 1 = 0.0512, w <i>R</i> 2 = 0.1057	
largest diff peak and hole	0.701 and -0.519 e Å ⁻³	0.981 and -1.503 e Å ⁻³	1.614 and -1.586 e Å ⁻³	0.582 and -0.314 e Å ⁻³	

^a [Ph₄P]₂[Cl₂Zn(μ -Cl)₂ZnCl₂] structure data (KAVHEU) acquired from the Cambridge Crystallographic Data Centre. Work found in Wilhelm, J. H.; Muller, U. Z. *Naturforsch., B: Chem. Sci.* **1989**, *44*, 1037.

temperature indicated in Table 1. Cell parameters were initially retrieved using COLLECT software¹⁴ and refined with HKL DENZO and SCALEPACK software, which was also used for data reduction.¹⁵ The structure was solved by direct methods using SHELXS-97 and refined by full-matrix least-squares method on *F*² with SHELXL-97.¹⁶

Single crystals of the three other compounds were mounted in thin-walled capillaries under a nitrogen atmosphere. The data were collected using the Bruker APEX2 software package¹⁷ on a Siemens diffractometer equipped with an Bruker APEXII CCD detector, a graphite monochromator, and Mo K α radiation ($\lambda = 0.71073$ Å). A hemisphere of data was collected in 1664 frames with 30 s exposure times. Data processing and absorption corrections were applied using APEX2 software. The structures were solved (direct methods) and refined using SHELX software.¹⁴ For all of the structures reported herein, the non-H atoms were refined anisotropically. Hydrogen atoms were placed in calculated positions using an appropriate riding model and coupled isotropic temperature factors.

Results and Discussion

All reactions were performed by mixing tetraphenylphosphonium halide (chloride and bromide) with diethylzinc in equimolar quantity, under strictly anhydrous conditions. The reactions are illustrated in Schemes 2 and 3. When the solvent (dichloromethane) was slowly evaporated over 12 h, crystals formed. In order to identify the products unambiguously, X-ray crystallographic studies were performed, and the materials were identified as [Ph₄P]₂[Et(Cl)Zn(μ -Cl)₂Zn-(Cl)Et] (yield = 22%) and [Ph₄P]₂[EtZnBr₃] (yield = 42%) (Figures 1 and 2). Elemental analysis data of the crystalline materials were in accordance with these structures. The analysis of [Ph₄P]₂[EtZnBr₃] indicated that no chloride ions were present (to detection limit). We also note that [Ph₄P]₂[EtZnBr₃] was found to be contaminated by a small amount of its non-ethylated analogue; the crystallographic analysis showed the anion to be disordered with a refined occupancy of 88% Et and 12% Br in one of the ligand positions. There are two unique anions in the crystal structure of [Ph₄P]₂[Et(Cl)Zn(μ -Cl)₂Zn(Cl)Et], one of which is significantly disordered. For the disordered anion (only half of which is crystallographically unique) both the bridging Cl and the terminal Et were found to be disordered over two

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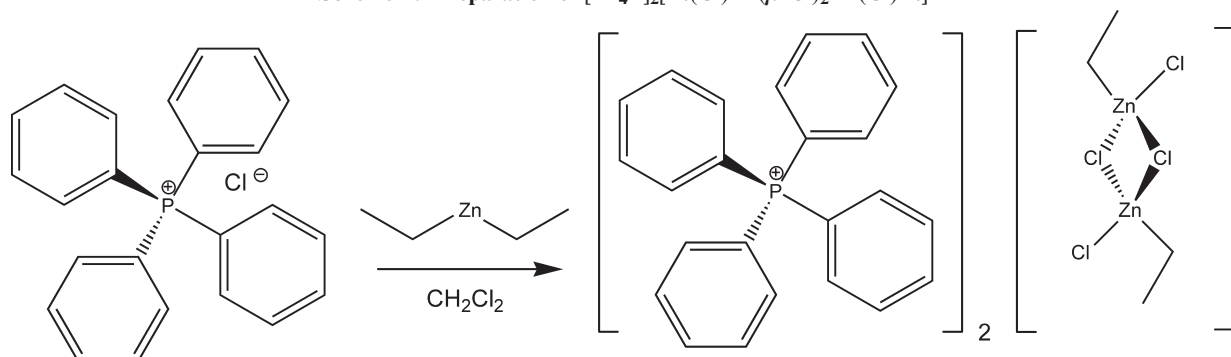
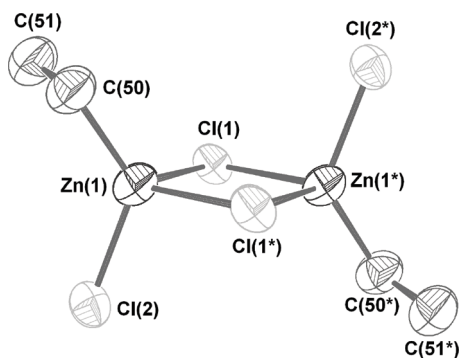
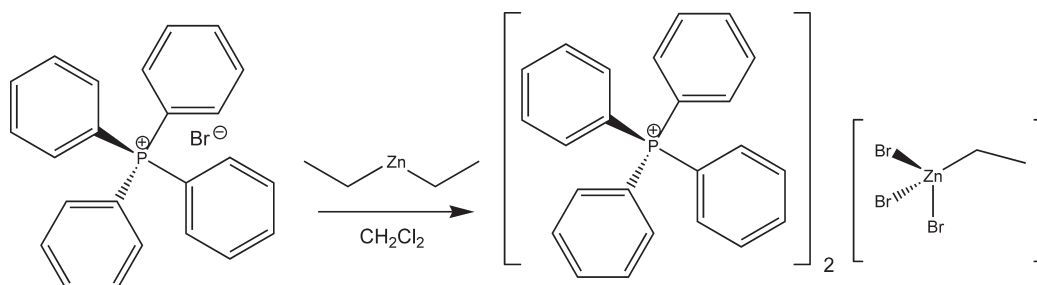
Scheme 2. Preparation of $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ Scheme 3. Preparation of $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$ 

Figure 1. ORTEP view of $[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$. Ellipsoids at the 50% level. Important distances (Å) and angles (deg): C(51)–C(50) 1.507(5), Zn(1)–C(50) 1.985(3), Zn(1)–Cl(2) 2.2675(8), Zn(1)–Cl(1*) 2.4771(7), Zn(1)–Cl(1) 2.4072(8); C(51)–C(50)–Zn(1) 114.7(2), C(50)–Zn(1)–Cl(2) 125.51(11), C(50)–Zn(1)–Cl(1*) 111.86(10), C(50)–Zn(1)–Cl(1) 116.86(11), Cl(2)–Zn(1)–Cl(1*) 104.66(3), Cl(2)–Zn(1)–Cl(1) 101.54(3), Cl(1)–Zn(1)–Cl(1*) 90.25(2), Zn(1)–Cl(1)–Zn(1*) 89.75(2). Atom symmetry transformation used to generate equivalent atoms: $-x+1, -y+2, -z+1$.

positions, while the final terminal position was found to be occupied by both Cl (58%) and Et (42%).

The solids isolated from the reaction mixtures were not highly soluble; however, proton NMR studies in CD_2Cl_2 solution indicated the presence of both aryl groups and ethyl fragments. The aromatic region in both cases displays three complex multiplets, which is typical of the tetraphenylphosphonium cation. Furthermore, a single ^{31}P NMR resonance was observed in each solution. The ^{13}C NMR spectra for these compounds were recorded, but were not sufficiently intense to observe the anion peaks.

The ^1H NMR spectrum of $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ in CD_2Cl_2 solution exhibited the anticipated triplet (1.05 ppm) and quartet (−0.11 ppm). For comparison, ZnEt_2

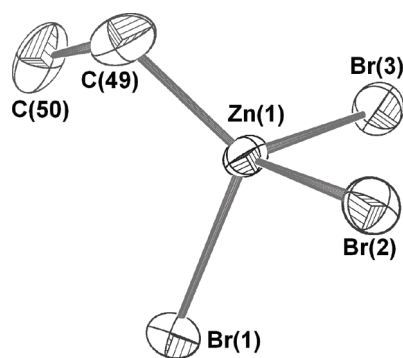


Figure 2. ORTEP view of EtZnBr_3^{2-} . Ellipsoids at 50% level. Important distances (Å) and angles (deg): C(49)–C(50) 1.588(17), Zn(1)–C(49) 2.008(13), Zn(1)–Br(1) 2.5097(8), Zn(1)–Br(2) 2.512(3), Zn(1)–Br(3) 2.4589(9); C(50)–C(49)–Zn(1) 110.5(8), C(49)–Zn(1)–Br(1) 112.7(4), C(49)–Zn(1)–Br(2) 114.2(3), C(49)–Zn(1)–Br(3) 115.0(4), Br(1)–Zn(1)–Br(2) 104.32(6), Br(3)–Zn(1)–Br(1) 105.65(3), Br(3)–Zn(1)–Br(2) 103.87(8).

in CD_2Cl_2 solution exhibits a triplet (1.05 ppm, CH_3) and quartet (0.20 ppm, CH_2). The significant upfield shift of the CH_2 relative to the parent compound was expected, as it has been observed in related studies.¹³ The ^1H NMR of $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$ was measured in DMSO, due to its insolubility in CD_2Cl_2 . This spectrum also consisted of the anticipated triplet and quartet, located at 1.03 and −0.20 ppm, respectively. ^1H and ^{13}C NMR spectra were also taken of the reaction media of the $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ and $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$ reactions, in an attempt to determine what other species were formed to account for the low yields and stoichiometry issues. Unfortunately, no evidence of other ethyl-containing compounds was seen, leaving the identity of the side products produced therein unknown.

Upon standing in solution for several days, reaction mixtures composed of tetraphenylphosphonium bromide and

diethylzinc yield $[\text{Ph}_4\text{P}]_2[\text{ZnBr}_4]$. We have isolated this compound from the reaction mixtures, and we have also prepared it by direct combination of ZnCl_2 or ZnBr_2 and the appropriate phosphonium salt. Specifically, we synthesized $[\text{Ph}_4\text{P}]_2[\text{ZnBr}_4]$ and $[\text{Ph}_4\text{P}]_2[\text{ZnCl}_4]$ purposely, by reaction of tetraphenylphosphonium halide and zinc(II) halide in THF. The cell parameters and melting points for $[\text{Ph}_4\text{P}]_2[\text{ZnBr}_4]$ and $[\text{Ph}_4\text{P}]_2[\text{ZnCl}_4]$ were clearly different from the alkyl derivatives reported above. For example, $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ crystallizes in the $P\bar{1}$ space group and $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$ in the $Pbca$ space group, while $[\text{Ph}_4\text{P}]_2[\text{ZnCl}_4]$ and $[\text{Ph}_4\text{P}]_2[\text{ZnBr}_4]$ both crystallize in the $C2/c$ space group. The melting points of these compounds were also found to be different from those observed for $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ and $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$. Lastly, the structures of $[\text{Zn}(\text{2,6-dimethoxypyridine})_4][\text{Zn}_2\text{Cl}_6]$ and $[\text{Ph}_4\text{P}][\text{Zn}_2\text{Br}_6]$ are known, and the data reported for these compounds are not similar to the crystallographic data reported herein for $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ or $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$.

The crystal structure of EtZnCl has been reported before.¹⁸ The structure of this material is significantly different than the structure found for $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ in the current paper, even though it can be formally considered the chloride complex of EtZnCl . For example, in the solid state EtZnCl exists as infinite puckered sheets composed of Zn_3Cl_3 six-membered rings. As anticipated, the terminal $\text{Zn}-\text{Cl}$ bond and two bridging $\text{Zn}-\text{Cl}$ bonds in EtZnCl (2.5408(7), 2.3486(6), and 2.5248(6) Å) are different from the corresponding terminal and bridging $\text{Zn}-\text{Cl}$ bonds in $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ (2.2675(8), 2.4072(8), and 2.4771(7) Å). The corresponding experimental $\text{C}-\text{C}$ bond length, on the other hand, is similar between the two species.

The structure of $[\text{Ph}_4\text{P}]_2[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]$ is closely related to one recently reported by Braustein,¹⁹ an organozinc analogue exhibiting a similar atom connectivity. Their structure, $\{(\text{PPh}_3\text{CH}_2)\text{ZnCl}_2\}_2$, also features a Zn_2Cl_2 heterocycle, but with neutral methylenetriphenylphosphorane groups replacing the formally anionic ethyl groups in $[\text{Et}(\text{Cl})\text{Zn}(\mu\text{-Cl})_2\text{Zn}(\text{Cl})\text{Et}]^{2-}$; thus $\{(\text{PPh}_3\text{CH}_2)\text{ZnCl}_2\}_2$ is a zwitterionic complex rather than a discrete anion/cation pair (i.e., a salt). The selected bond lengths of this structure, $\text{Zn}-\text{C}$ 2.025(2) Å, $\text{Zn}-\text{Cl1}$ 2.3505(6) Å, and $\text{Zn}-\text{Cl2}$ 2.2491(7) Å, highlight the marked difference between the structures, and the relatively longer $\text{Zn}-\text{C}$ bond suggests a more loosely attached carbon.

Parenthetically, we originally attempted to prepare similar complexes using tetrabutylphosphonium halide salts as the source of halide. These were chosen because their bulk and the valence saturation of the heteroatom of the cations made it less likely that the cations would participate in the reaction. The product mixtures did not yield crystalline solids but gave instead reactive oils. Upon noting this, the butyl substituents were swapped for phenyl groups, in order to increase the symmetry of the structure, which resulted in the crystalline products reported. Formation of oils (ionic liquids?) as a dense liquid phase was also reported by Richey in the reaction of ammonium halides with dialkylzinc reagents.¹⁰

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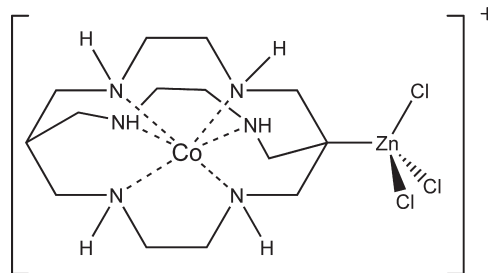


Figure 3. Hexaazabicyclo[6.6.6]icosane cobalt(III), a rare organozinc trihalide complex.

Perhaps the most intriguing result of this study was the isolation of $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$. Trihaloalkylzinc compounds are an exceedingly rare structural motif, with only one related structural confirmation, specifically, the organozinc trihalide seen in Figure 3.²⁰

This compound, which consists of a cobalt(III) ion caged in a zinc trichloride-capped hexamine ligand, shows zwitterionic stabilization reminiscent of that in $\{(\text{PPh}_3\text{CH}_2)\text{ZnCl}_2\}_2$. Like $\{(\text{PPh}_3\text{CH}_2)\text{ZnCl}_2\}_2$, the positively charged alkyl fragment helps to balance the zinc's formal anionic charge, leaving a stable organozinc moiety. We feel it is important to note that the charge separation in $\{(\text{PPh}_3\text{CH}_2)\text{ZnCl}_2\}_2$ and the hexaazabicyclo[6.6.6]icosane cobalt(III) complex is fundamentally different from that of $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$, which instead exists as discrete ions. This likely affects the nature of the $\text{Zn}-\text{C}$ bond. It is supposed that the formation of the $[\text{Ph}_4\text{P}]_2[\text{EtZnBr}_3]$ structure was, in part, induced by the high concentration of charged species in the evaporating solvent.

We next decided to preform a computational study of the stepwise addition of Br^- to $[\text{EtZn}]^+$, in which we calculated the energy change for each step (Table 2). At the density functional B3LYP/6-31+G* level of calculation, the energy changes found for the first two additions are negative, but the final addition instead gives a higher energy product. These results indicate that the addition of Br^- to $[\text{EtZnBr}_2]^-$ is not energetically favored. The dianionic crystal is likely formed due to the stabilization effect of the crystal lattice energy. However, the energy disparity suggests that, in solution, the Br^- may leave the dianionic complex, potentially allowing other reactions to occur. The effects of solvation were thus calculated, using gas-phase geometry, with CH_2Cl_2 as the solvent. The *ab initio* molecular orbital calculations were carried out with Gaussian 03.²¹ Geometry optimizations, followed by harmonic vibrational frequency calculations to characterize the stationary points,

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Table 2. Energetics (kJ/mol) of Stepwise Addition of Br[−] to [EtZn]⁺ ^a

reaction	B3LYP/6-31+G*	CPCM/B3LYP/6-31+G*	CPCM/B3LYP/6-31+G* + non-electrostatic term
EtZn ⁺ + Br [−] → EtZnBr	−739.8	−102.5	−103.5
EtZnBr + Br [−] → EtZnBr ₂ [−]	−159.4	−63.7	−62.0
EtZnBr ₂ [−] + Br [−] → EtZnBr ₃ ^{2−}	192.2	−2.7	1.8

^a The second column gives the electronic-only gas-phase internal energy calculation, the third column represents the free energy of reaction using an electrostatic dielectric continuum, and the final column also includes non-electrostatic terms such as cavitation energy.

were sequentially performed at the HF/STO-3G, HF/3-21G, HF/6-31G(d), HF/6-31+G(d), and B3LYP/6-31+G(d) levels of theory, followed by a single-point calculation at the CPCM/B3LYP/6-31+G* level. Only the higher level results are reported. The energy difference in the final Br[−] addition was found to be only −2.7 kJ/mol. Because this transition is nearly thermoneutral, it appears that a dynamic equilibrium between the EtZnBr₂[−] and EtZnBr₃^{2−} species likely exists in solution. This equilibrium has been noted in previous studies.²²

In summary, we have isolated two novel organozinc species whose formation is likely due to the ionic strength of the solution in which they formed. We admit that we are unsure of the fate of the remaining Zn-ethyl groups in these mixtures, although it is possible that they form [R₃Zn] species, as suggested by previous studies.¹³ The use of salts with unreactive cations or anions, such as the tetraphenylphosphonium halide salts used herein, suggests possible explanations for some of the exotic chemistry often seen in organozinc systems involving halide ions.¹³ Specifically, the chemistry observed when

utilizing active organometallic species (such as Et₂Zn) in ionic liquids, as seen in the results of Law, Wong, and Chan,⁸ suggests that the halide ions in solution play a critical role. They determine the reactivity of the added reagents and increase the potential for formation of ionic complexes, as described in this paper.

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Supporting Information Available: Crystallographic information files (CIF) have been deposited with the CCDC, 12 Union Road, Cambridge, CB2 1EZ, U.K. These can be obtained on request free of charge, by quoting the publication citation and deposition numbers 750430–750433. This material is also available free of charge via the Internet at <http://pubs.acs.org>.

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